

## Where DGLAP comes from in Row IV

The  $p^+$  bracket is  $P_{gg}$ : the plus prescription, the fragmentation measure, and the collinear pole

$$\zeta = \frac{k^+}{k^+ + p^+}, \quad \bar{\zeta} = 1 - \zeta = \frac{p^+}{k^+ + p^+}, \quad r = y' - y, \quad \kappa = k_\perp \cdot r, \quad X = \frac{x - y}{(x - y)^2}, \quad X' = \frac{x' - y'}{(x' - y')^2}, \quad \mathcal{M} = \frac{1}{\epsilon} + 1 - \log(4\pi^2 \mu_{\text{MS}}^2 r^2)$$

**The one line that does it.** The  $\mathfrak{C}_1$  vertex splits a parent of  $S = p^+ + k^+$  into the measured gluon ( $k^+$ ) and an unmeasured one ( $p^+$ ). Call the measured fraction  $\zeta = k^+/S$ . Then

$$\frac{p^+}{(k^+ + p^+)^2} + \frac{p^+}{k^+} + \frac{1}{p^+} = \frac{1}{k^+} \left[ \zeta \bar{\zeta} + \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right] = \frac{P_{gg}(\zeta)}{2N_c k^+}$$

**exactly** — your three  $p^+$  structures are the three terms of the gluon splitting function, term by term. Everything below follows from that.

### 1. The three signatures of fragmentation, all present

**(i) The measure.**  $p^+ = k^+ \frac{1-\zeta}{\zeta} \Rightarrow dp^+ = -\frac{k^+}{\zeta^2} d\zeta$ , so

$$\int_\Lambda^{\vee-k^+} dp^+ = k^+ \int_{\zeta_0}^{\zeta_1} \frac{d\zeta}{\zeta^2}, \quad \zeta_0 = \frac{k^+}{\vee}, \quad \zeta_1 = \frac{k^+}{k^+ + \Lambda} \simeq 1 - \frac{\Lambda}{k^+}$$

$\frac{d\zeta}{\zeta^2}$  is the fragmentation measure, not  $d\zeta$ .

**(ii) The hard scale.** Your phase is  $e^{-ik \cdot r} e^{-ik \cdot r p^+/k^+} = e^{-ik \cdot r \frac{S}{k^+}}$  and  $\frac{S}{k^+} = \frac{1}{\zeta}$ , so

$$e^{-i \frac{k_\perp}{\zeta} \cdot (y' - y)}$$

— the hard factor is evaluated at  $k_\perp/\zeta$ , not at  $k_\perp$ . This is the single most characteristic fragmentation signature, and it is exactly the  $\frac{S}{k^+}$  that the  $w, w'$  delta functions put into the exponent.

**(iii) Where the Wilson lines sit.** At  $x, x', y, y'$  — the *parent* positions, with no  $\zeta$  dependence at all. The splitting is a final-state one, after the target has been traversed. If the Wilson lines depended on  $\zeta$  there would be no factorisation.

### 2. The row in $\zeta$ variables

Starting from the (corrected) three-block form and substituting  $\frac{p^+}{S^2} = \frac{\zeta \bar{\zeta}}{k^+}, \frac{p^+}{k^+} + \frac{1}{p^+} = \frac{1}{k^+} \left[ \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right]$ , with  $\frac{k^+}{2\pi} \cdot \pi \cdot \frac{1}{k^+} = \frac{1}{2}$ :

$$\begin{aligned} \frac{d^3 N^{\text{NLO}}}{d^3 k} &= \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int_{\zeta_0}^{\zeta_1} \frac{d\zeta}{\zeta^2} \int_{x, x', y, y'} e^{-i \frac{k_\perp}{\zeta} \cdot (y' - y)} \left\{ \frac{P_{gg}(\zeta)}{2} \mathcal{M}(X \cdot X') \right. \\ &\quad \left. - N_c \left[ d_\perp \zeta \bar{\zeta} (X \cdot \hat{r})(X' \cdot \hat{r}) + \left( \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right) (X \cdot X') \right] \right\} \times \mathcal{U} \end{aligned}$$

with  $\mathcal{U} = [U^{e'c}(y') - U^{e'c}(x')] [U^{ec}(y) - U^{ec}(x)] \rho^{e'}(x') \rho^e(x)$  and  $P_{gg}(\zeta) = 2N_c \left[ \zeta \bar{\zeta} + \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right]$ .

### 3. The plus prescription is the DGLAP one

$p^+ \rightarrow 0$  is  $\zeta \rightarrow 1$ , and the pole there is the  $\frac{\zeta}{1-\zeta}$  of  $P_{gg}$  — the standard DGLAP endpoint. The test is whether the companion is regular there. Collecting everything that multiplies  $\frac{1}{1-\zeta}$  in the integrand,

$$\frac{1}{\zeta^2} e^{-i \frac{k_\perp}{\zeta} \cdot (y' - y)} \frac{\zeta}{1-\zeta} [\mathcal{M} - 1](X \cdot X') = \frac{h(\zeta)}{1-\zeta} [\mathcal{M} - 1](X \cdot X'), \quad h(\zeta) = \frac{e^{-i \kappa / \zeta}}{\zeta}$$

and  $h(1) = e^{-i \kappa}$  is finite.  $\mathcal{M}, X, X', \hat{r}$  and the Wilson lines carry no  $\zeta$ . So the prescription applies with nothing extra:

$$\begin{aligned} \frac{1}{1-\zeta} &\longrightarrow \left[ \frac{1}{1-\zeta} \right]_+ + \delta(1-\zeta) \log \frac{k^+}{\Lambda}, & \int_0^1 d\zeta \left[ \frac{1}{1-\zeta} \right]_+ f(\zeta) &\equiv \int_0^1 d\zeta \frac{f(\zeta) - f(1)}{1-\zeta} \\ &\implies \frac{P_{gg}(\zeta)}{2N_c} \longrightarrow \underbrace{\zeta \bar{\zeta} + \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{[1-\zeta]_+}}_{\text{regularised splitting function}} + \delta(1-\zeta) \log \frac{k^+}{\Lambda} \end{aligned}$$

Numerical confirmation of the split (direct vs plus form), for  $\int_{\zeta_0}^{1-\lambda} \frac{d\zeta}{\zeta(1-\zeta)} e^{-i\kappa/\zeta}$ :

| $\kappa$ | $\lambda$ | $\zeta_0$ | direct                    | plus form                 |
|----------|-----------|-----------|---------------------------|---------------------------|
| 0.7      | $10^{-6}$ | 0.02      | $9.374373 - 9.986195 i$   | $9.374374 - 9.986196 i$   |
| 2.0      | $10^{-7}$ | 0.05      | $-7.587117 - 12.864336 i$ | $-7.587117 - 12.864336 i$ |

**Reading the two pieces.** The  $\delta(1-\zeta) \log \frac{k^+}{\Lambda}$  is the rapidity divergence — a virtual-like,  $\zeta$ -integrated term that goes with the JIMWLK evolution of  $[U(y') - U(x')][U(y) - U(x)]$ . The  $[\frac{1}{1-\zeta}]_+$  part is genuine  $\zeta < 1$  splitting and is what DGLAP acts on. In the complete  $P_{gg}$  the  $\delta(1-\zeta)$  coefficient is  $\frac{\beta_0}{2} = \frac{11N_c}{6}$  (pure glue), built from the virtual rows; here you get  $\log \frac{k^+}{\Lambda}$  instead, because this single row has no virtual partner. That coefficient is only assembled once all of Group II is summed.

#### 4. The collinear pole, in collinear-factorisation form

Define the LO structure (the only  $\zeta$ -dependence is through the hard scale)

$$\mathcal{N}_{\text{LO}}(q_\perp) \equiv \int_{x, x', y, y'} e^{-iq_\perp \cdot (y' - y)} \frac{(x - y) \cdot (x' - y')}{(x - y)^2 (x' - y')^2} \mathcal{U}$$

Then the  $\frac{1}{\epsilon}$  part of the row is, with no approximation,

$$\left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{pole}} = \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \frac{1}{\epsilon} \int_{\zeta_0}^{\zeta_1} \frac{d\zeta}{\zeta^2} \frac{P_{gg}(\zeta)}{2} \mathcal{N}_{\text{LO}}\left(\frac{k_\perp}{\zeta}\right)$$

which is exactly the shape of collinear factorisation for fragmentation,

$$\frac{d\sigma^h}{d^3 k} = \int \frac{d\zeta}{\zeta^2} D_{h/g}(\zeta, \mu_F) \frac{d\hat{\sigma}}{d^3 k}\left(\frac{k_\perp}{\zeta}\right)$$

with the  $\frac{1}{\epsilon}$  removed by the bare-to-renormalised relation

$$D_{g/g}(\zeta, \mu_F) = \delta(1-\zeta) + \frac{\alpha_s}{2\pi} \left[ -\frac{1}{\epsilon} + \log \frac{\mu_F^2}{\mu^2} \right] P_{gg}(\zeta) + O(\alpha_s^2)$$

#### The normalisation check

With the LO written in the same convention,  $\frac{2}{(2\pi)^3} |\mathfrak{A}^{(1)}|^2 = \frac{1}{(2\pi)^3} \frac{g^2}{\pi^2 k^+} \mathcal{N}_{\text{LO}}$ , the ratio of coefficients is

$$\frac{g^4/(8\pi^4)}{g^2/\pi^2} = \frac{g^2}{8\pi^2} = \frac{\alpha_s}{2\pi}, \quad \implies \quad \left. \frac{d^3 N^{\text{NLO}}}{d^3 k} \right|_{\text{pole}} = \frac{\alpha_s}{2\pi} \frac{1}{\epsilon} \int \frac{d\zeta}{\zeta^2} \frac{P_{gg}(\zeta)}{2} \frac{d^3 N^{\text{LO}}}{d^3 k}\left(\frac{k_\perp}{\zeta}\right)$$

i.e. the textbook  $\frac{\alpha_s}{2\pi\epsilon} P_{gg}$ , up to the leftover  $\frac{1}{2}$ .

That residual  $\frac{1}{2}$  is **not** something I can settle from this row alone: it depends on how you normalise the LO ( $\frac{2}{(2\pi)^3}$  vs  $\frac{4}{(2\pi)^3}$ , the same ambiguity as the  $\frac{4}{(2\pi)^3}$  in your (0.1)) and on whether the hermitian-conjugate row  $\bar{\mathfrak{C}}\mathfrak{A}^{(1)} \times \mathfrak{A}^{(1)\dagger}\bar{\mathfrak{C}}^\dagger$  is counted separately or already included. Fix it against your own LO expression — the  $\zeta$  structure, which is what matters for DGLAP, is exact either way.

## 5. What the $\frac{1}{\epsilon}$ actually is — and a correction to what I said before

Earlier I read the pole of  $\mathcal{G}^{mm'}$  as an infrared one to be cancelled against the Group II rows in which  $U(z)$  survives. The  $\zeta$  rewriting shows the correct home. The pole comes from  $|z| \rightarrow \infty$  in  $\int d^2 z \frac{(y-z)^m (y'-z)^{m'}}{(y-z)^2 (y'-z)^2}$ , i.e. from small transverse momentum of the emitted gluon relative to its parent — that is the **final-state collinear region** of the  $g \rightarrow gg$  splitting. Its residue is  $P_{gg}(\zeta)$  with the  $\frac{d\zeta}{\zeta^2}$  measure and the hard factor at  $k_\perp/\zeta$ , which is the definition of a fragmentation contribution. So it is absorbed into the gluon fragmentation function at the scale  $\mu_F$ , not cancelled. The statement that survives from before is only the narrower one: the  $\delta(1 - \zeta)$  piece, carrying  $\log \frac{k^+}{\Lambda}$ , still has to be combined with the other rows before it can be handed to the evolution.

## 6. What is left after the subtraction

Once  $\frac{1}{\epsilon} P_{gg}$  is absorbed into  $D_{g/g}(\zeta, \mu_F)$ , the finite remainder of this row is the NLO coefficient function:

$$\begin{aligned} & \frac{1}{(2\pi)^3} \frac{g^4}{8\pi^4} \frac{1}{k^+} \int \frac{d\zeta}{\zeta^2} \int_{x, x', y, y'} e^{-i \frac{k_\perp}{\zeta} \cdot r} \left\{ \frac{P_{gg}(\zeta)}{2} \left[ 1 - \log(4\pi^2 \mu_F^2 r^2) \right] (X \cdot X') \right. \\ & \left. - N_c \left[ d_\perp \zeta \bar{\zeta} (X \cdot \hat{r})(X' \cdot \hat{r}) + \left( \frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}} \right) (X \cdot X') \right] \right\} \mathcal{U} \end{aligned}$$

Three remarks on it:

- The  $\log(4\pi^2 \mu_F^2 r^2)$  sits under the  $x, x', y, y'$  integrals, so it does **not** factor out of  $\mathcal{N}_{\text{LO}}$ . That is the usual statement that the coefficient function is not a pure number: the position-space log is the NLO hard factor.
- The  $d_\perp \zeta \bar{\zeta} (X \cdot \hat{r})(X' \cdot \hat{r})$  term is the azimuthal partner of the  $\zeta \bar{\zeta}$  piece of  $P_{gg}$ . It carries no pole and no  $\frac{1}{1-\zeta}$ , so it neither fragments nor needs the plus prescription.
- The  $-\left(\frac{\bar{\zeta}}{\zeta} + \frac{\zeta}{\bar{\zeta}}\right)(X \cdot X')$  term does have the  $\zeta \rightarrow 1$  pole, and it is what turns  $\mathcal{M}$  into  $\mathcal{M} - 1$  there — so the  $[\ ]_+$  acts on the *sum* of it and the  $P_{gg}\mathcal{M}$  term, with the finite residue  $(\mathcal{M} - 1)(X \cdot X')$  quoted in §3.

## 7. Two loose ends that affect the DGLAP reading

- **The small- $\zeta$  end.**  $\zeta \rightarrow \zeta_0 = \frac{k^+}{\sqrt{}}$  is the measured gluon carrying a vanishing fraction. There  $P_{gg} \rightarrow \frac{2N_c}{\zeta}$  and the measure is  $\frac{d\zeta}{\zeta^2}$ , so the integrand grows as  $\zeta^{-3}$ . What controls it here is the phase  $e^{-ik_\perp \cdot r/\zeta}$ , which oscillates ever faster as  $\zeta \rightarrow 0$ ; in a genuine FF treatment it is the small- $z$  behaviour of  $D_{g/g}$ . Either way, do not send  $\vee \rightarrow \infty$  before the  $y, y'$  integrals.
- **Where  $\Lambda$  comes from.** The  $\Theta(k_{\text{arg}}^+ - p_{\text{arg}}^+ - \Lambda)$  in your  $\mathfrak{E}_{1jki}$  reduces to  $\Theta(k^+ - \Lambda)$  in this row and therefore does **not** enforce  $\zeta < 1 - \frac{\Lambda}{k^+}$ . The endpoint that the plus prescription regulates needs its own  $\Theta(p^+ - \Lambda)$ . Worth settling before quoting the  $\log \frac{k^+}{\Lambda}$  coefficient.

## 8. Summary

| question                                      | answer                                                                                                                                                                                                         |
|-----------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Can you use the plus prescription for $p^+$ ? | <b>Yes.</b> In $\zeta = \frac{k^+}{k^+ + p^+}$ it is the standard DGLAP $[\frac{1}{1-\zeta}]_+$ ; the companion $h(\zeta) = \frac{e^{-i\kappa/\zeta}}{\zeta}$ is regular at $\zeta = 1$ .                      |
| Does DGLAP appear?                            | <b>Yes.</b> The $p^+$ bracket is exactly $\frac{P_{gg}(\zeta)}{2N_c k^+}$ , the measure is $\frac{d\zeta}{\zeta^2}$ , the hard factor sits at $\frac{k}{\zeta}$ and the Wilson lines are $\zeta$ -independent. |
| What absorbs the $\frac{1}{\epsilon}$ ?       | The gluon fragmentation function $D_{g/g}(\zeta, \mu_F)$ — this is a final-state collinear pole, not a charge renormalisation and not a cancellation.                                                          |
| What carries $\Lambda$ ?                      | The $\delta(1 - \zeta)$ term, $\log \frac{k^+}{\Lambda}$ — the rapidity log, for JIMWLK.                                                                                                                       |

Verified in `src/rowG2_dglap.py`: the identity  $\frac{p^+}{s^2} + \frac{p^+}{k^+} + \frac{1}{p^+} = \frac{P_{gg}(\zeta)}{2N_c k^+}$  (difference exactly 0), its consistency with the  $\sigma = 1/\zeta$  masters  $J_1 + J_2$ , the plus-prescription split against direct quadrature, and the  $\frac{\alpha_s}{2\pi}$  normalisation.